



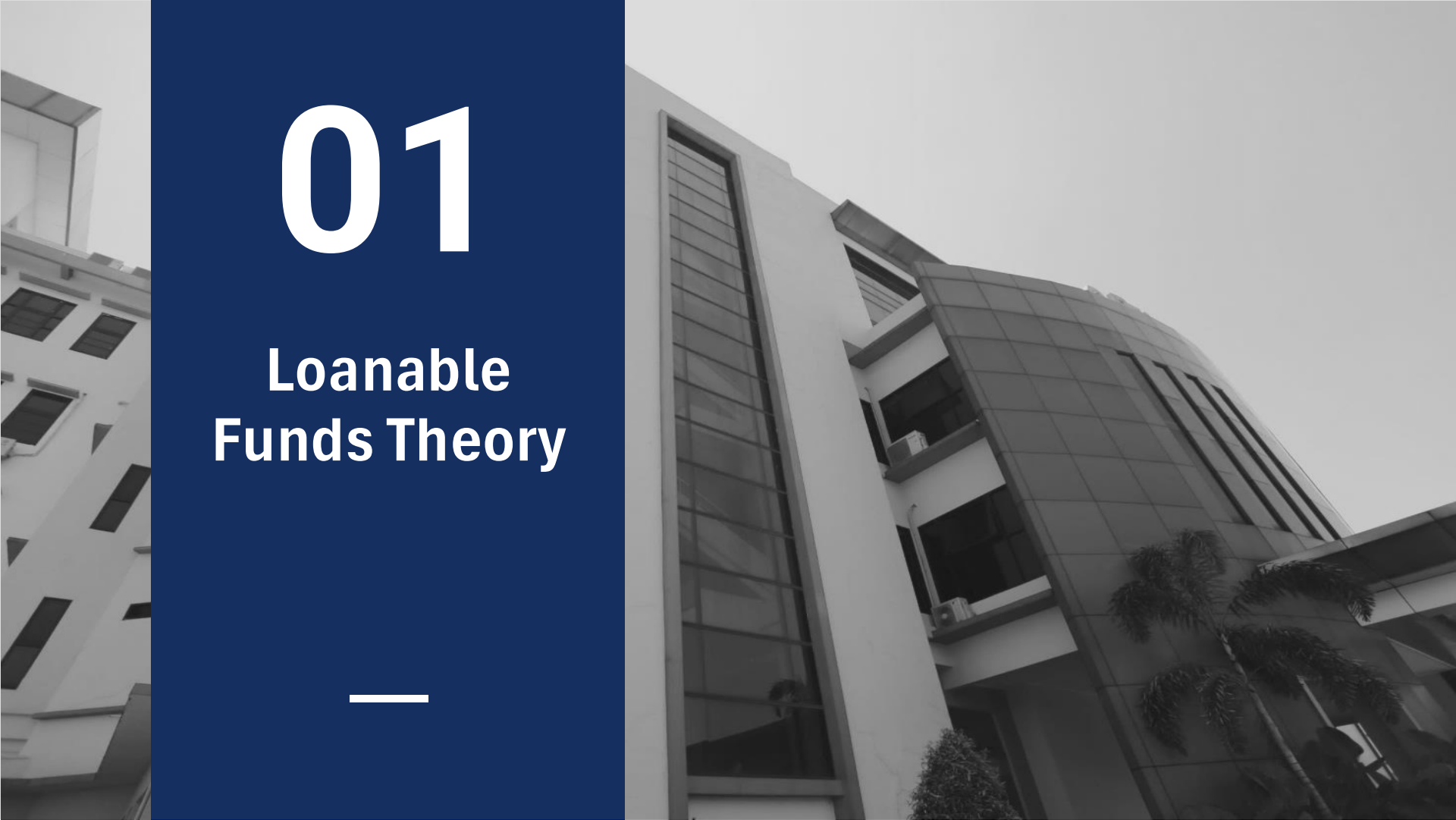
Determination of Interest Rates, and Structure of Interest Rates

Team Teaching Financial Markets and Institutions



Contents of This Template

1. Loanable Funds Theory (2-1)
2. Factors That Affect Interest Rates (2-2)
3. Why Debt Security Yields Vary (3-1)
4. Modeling the Yield to be Offered on a Debt Security (3-2)
5. A Closer Look at the Term Structure (3-3)



01

Loanable Funds Theory



Loanable Funds Theory

The loanable funds theory explains interest rate movements based on the supply and demand for loanable funds. The demand side includes borrowing by households, businesses, and governments, while the supply side represents funds provided by savers. Equilibrium is achieved when demand equals supply, determining the market interest rate.

Household demand is influenced by housing expenditures and installment debt, business demand by investment in long-term assets, and government demand by budget deficits. Foreign demand is affected by the difference between domestic and U.S. interest rates.

Demand Factors

- Household spending
- Business investment
- Government deficits
- Foreign borrowing

Supply Factors

- Household savings
- Business cash flows
- Government revenues
- Foreign investment



Household & Business Demand

Households demand loanable funds for housing, auto purchases, and other items, with demand inversely related to interest rates. Businesses need funds for long-term investments, increasing demand when rates are low. Government demand, particularly the federal government, is often interest-inelastic.

Foreign demand is influenced by the difference between domestic and U.S. interest rates. The aggregate demand for loanable funds is the sum of demands from all sectors, inversely related to the interest rate.

Household

Driven by spending needs and income levels.

Business

Driven by investment opportunities and project costs.

Government

Driven by fiscal policy and budget deficits.





Supply of Loanable Funds

The supply of loanable funds comes from savers, primarily households, and is positively related to interest rates. Foreign entities also supply funds by purchasing domestic securities. The Federal Reserve influences the supply through monetary policy, affecting interest rates and borrowing.

The aggregate supply schedule combines all sector supply schedules with the Fed's monetary policy. Shifts in the supply curve occur with changes in tax rates on interest income.

1

Households

Primary suppliers through savings.

2

Foreign Entities

Significant suppliers via security purchases.

3

Federal Reserve

Influences supply via monetary policy.

Equilibrium Interest Rate

The equilibrium interest rate equates the aggregate demand and supply of loanable funds. Algebraically, this occurs when $DA = SA$. Graphically, it's the intersection of the demand and supply curves. Changes in demand or supply disrupt equilibrium, leading to interest rate adjustments.

For example, increased borrowing desire raises DA, causing a shortage and higher rates. Conversely, increased saving raises SA, leading to a surplus and lower rates.

1

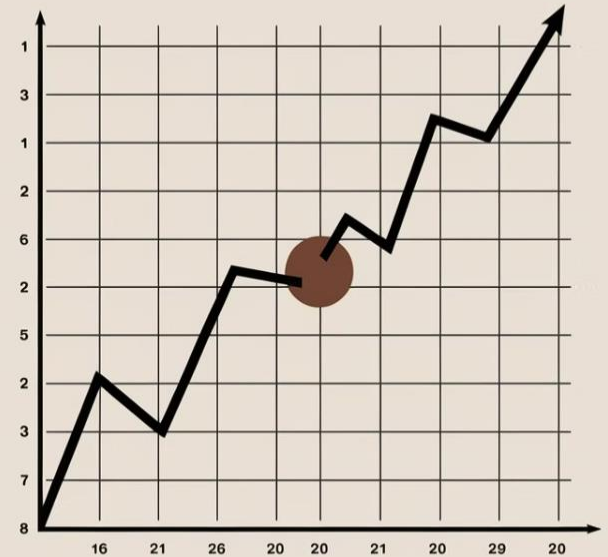
Demand Increase

Shortage of funds, rates rise.

2

Supply Increase

Surplus of funds, rates fall.





The equilibrium interest rate is the rate that equates the aggregate demand for funds (aggregate quantity of funds demanded) with the aggregate supply of loanable funds. The aggregate demand for funds () D_A can be written as

$$D_A = D_h + D_b + D_g + D_m + D_f$$

where

D_h = household demand for loanable funds

D_b = business demand for loanable funds

D_g = federal government demand for loanable funds

D_m = municipal government demand for loanable funds

D_f = foreign demand for loanable funds

The aggregate supply of funds (S_A) can likewise be written as

$$S_A = S_h + S_b + S_g + S_m + S_f$$

where

S_h = household supply of loanable funds

S_b = business supply of loanable funds

S_g = federal government supply of loanable funds

S_m = municipal government supply of loanable funds

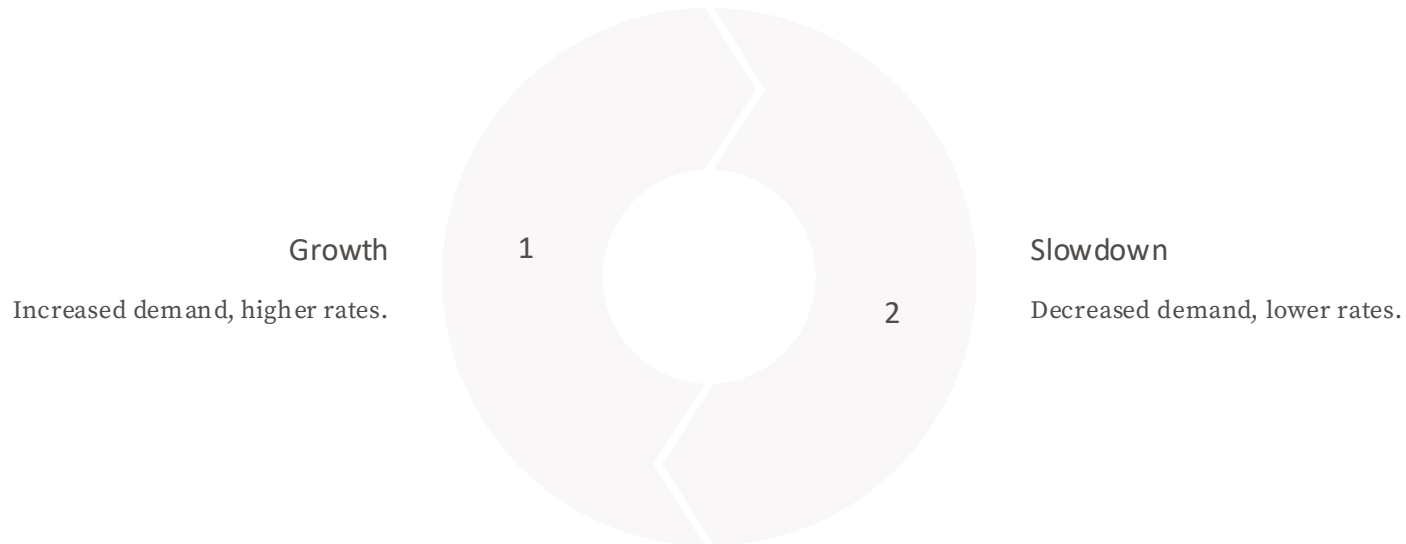
S_f = foreign supply of loanable funds



Impact of Economic Growth

Economic growth shifts the demand curve for loanable funds, affecting equilibrium interest rates. Favorable conditions increase business cash flows, making more projects acceptable and increasing demand. This outward shift in the demand curve raises the equilibrium interest rate.

Conversely, an economic slowdown shifts the demand curve inward, decreasing the equilibrium interest rate. The supply curve may shift slightly, but the demand shift is more significant.

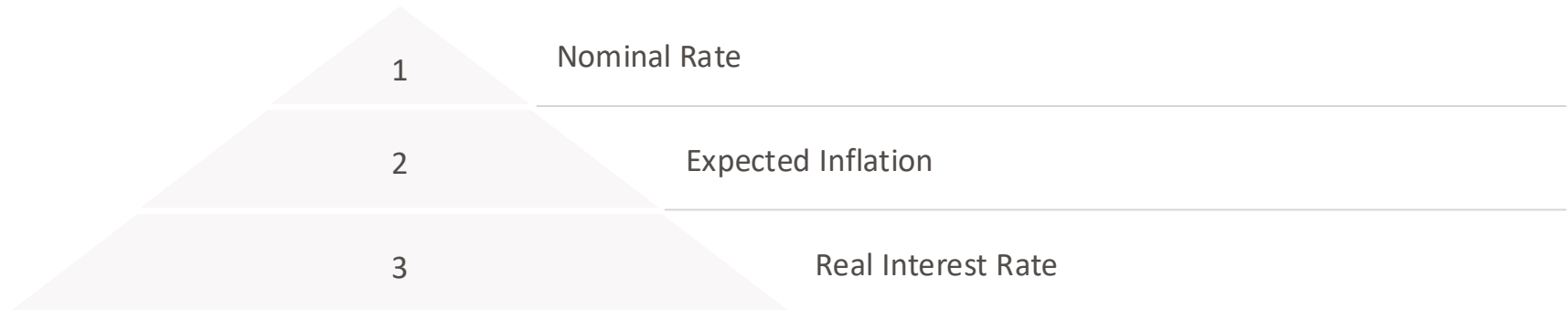




Impact of Inflation

Inflationary expectations affect interest rates by altering spending behavior. Higher inflation expectations reduce savings and increase borrowing, shifting the supply curve inward and the demand curve outward. This results in a higher equilibrium interest rate.

The Fisher effect proposes that the nominal interest rate compensates savers for reduced purchasing power due to inflation and provides an additional premium for forgoing consumption.





Fisher Effect

$$i = E(\text{INF}) + i_R$$

Where:

i = nominal rate of interest

$E(\text{INF})$ = expected inflation rate

i_R = real interest rate

The difference between the nominal interest rate and the expected inflation rate is referred to as the real interest rate because it measures the rate of interest earned by a saver after adjusting for the expected loss in purchasing power (due to expected inflation) over the time period of concern.

The preceding equation can be rearranged to express the real interest rate as

$$i_R = i - E(\text{INF})$$

To the extent that the real rate of interest is very stable over time, the Fisher effect suggests that the nominal interest rate should rise in response to an increase in expected inflation.

02

Factors That Affect Interest Rates





Impact of Monetary Policy

The Federal Reserve influences interest rates by adjusting the money supply. Increasing the money supply lowers interest rates, encouraging borrowing and spending. Reducing the money supply raises interest rates.

During weak economic conditions, the Fed may increase the money supply to stimulate the economy. Conversely, to combat inflation, the Fed may reduce the money supply.

1

Increase Money Supply

2

Lower Interest Rates

3

Stimulate Economy



Impact of Budget Deficit

Federal government budget deficits increase the demand for loanable funds, shifting the demand curve outward. This can lead to higher interest rates and the crowding-out effect, where government borrowing reduces private sector borrowing.

Excessive government demand for funds can drive up interest rates, making it more expensive for consumers and corporations to borrow

03

Why Debt Security Yields Vary



Why Understanding Yield Variation Matters



Individual and institutional investors must understand why quoted yields vary to determine whether extra yield on a security outweighs unfavorable characteristics.

Financial managers of corporations or government agencies need this knowledge to estimate the yield they must offer to successfully sell new debt securities.

- ❏ Even a 1% difference in yield on \$30 million in bonds equals \$300,000 in additional annual interest expenses.



Four Key Characteristics That Drive Yields



Credit Risk

The risk of default by the issuer. Securities with higher default risk must offer higher yields to attract investors.



Liquidity

How easily a security can be converted to cash without loss of value. Less liquid securities require higher yields.



Tax Status

Whether interest is taxable or tax-exempt. Taxable securities must offer higher before-tax yields to compete.



Term to Maturity

The length of time until the security matures. Yield varies based on the relationship between term and risk.



Credit Risk and the Premium Investors Demand

Most securities carry default risk, so investors must assess the creditworthiness of issuers. Securities with higher credit risk must offer a credit risk premium—a higher yield above Treasury bonds—to attract investors.

If all other characteristics are equal, securities with unfavorable characteristics must offer higher yields to entice investors to buy them.

Treasury Bonds

7% yield, risk-free baseline

Zanstell Co. Bonds

8% yield, 1% credit risk premium



Bond Rating Systems

Rating agencies like Moody's and Standard & Poor's assess creditworthiness of bond issuers. These ratings help investors evaluate credit risk without conducting their own detailed analysis.

Commercial banks typically invest only in investment-grade bonds (Baa/BBB or better). The credit risk premium varies significantly: 1% on highly rated bonds, 2.5% on medium-quality bonds, and 5% on low-quality bonds.

Investment Grade Threshold

Moody's: Baa or better

S&P: BBB or better

These ratings separate institutional-quality bonds from speculative investments.

Bond Rating Classifications

Description	Moody's	Standard & Poor's
Highest quality	Aaa	AAA
High quality	Aa	AA
High-medium quality	A	A
Medium quality	Baa	BBB
Medium-low quality	Ba	BB
Low quality (speculative)	B	B
Poor quality	Caa	CCC
Very poor quality	Ca	CC
Lowest quality (in default)	C	D

Rating agencies charge issuers fees for assessments. Ratings can change over time based on financial conditions and economic environment. During recessions, credit risk premiums increase significantly as default concerns rise.

Liquidity and Tax Considerations

Liquidity Premium

Investors prefer securities easily converted to cash without value loss. Securities with short-term maturity or active secondary markets have greater liquidity. Less liquid securities must offer higher yields to compensate investors.

Tax Status Impact

After-tax income matters most to investors. Treasury interest is federally taxable but state-exempt. Municipal securities are often federally and state tax-exempt. Corporate debt is fully taxable. The formula: $Y_{at} = Y_{bt} (1 - T)$

$$Y_{at} = Y_{bt} (1 - T)$$

where

Y_{at} = after-tax yield

Y_{bt} = before-tax yield

T = investor's marginal tax rate

Investors retain only a percentage $(1 - T)$ of the before-tax yield once taxes are paid.

Example

Consider a taxable security that offers a before-tax yield of 8 percent. When converted to after-tax terms, the yield will be reduced by the tax percentage. The precise after-tax yield depends on the tax rate T . If the investor's tax rate is 20 percent, then the after-tax yield will be

$$\begin{aligned} Y_{at} &= Y_{bt}(1 - T) \\ &= 8\%(1 - 0.2) \\ &= 6.4\% \end{aligned}$$

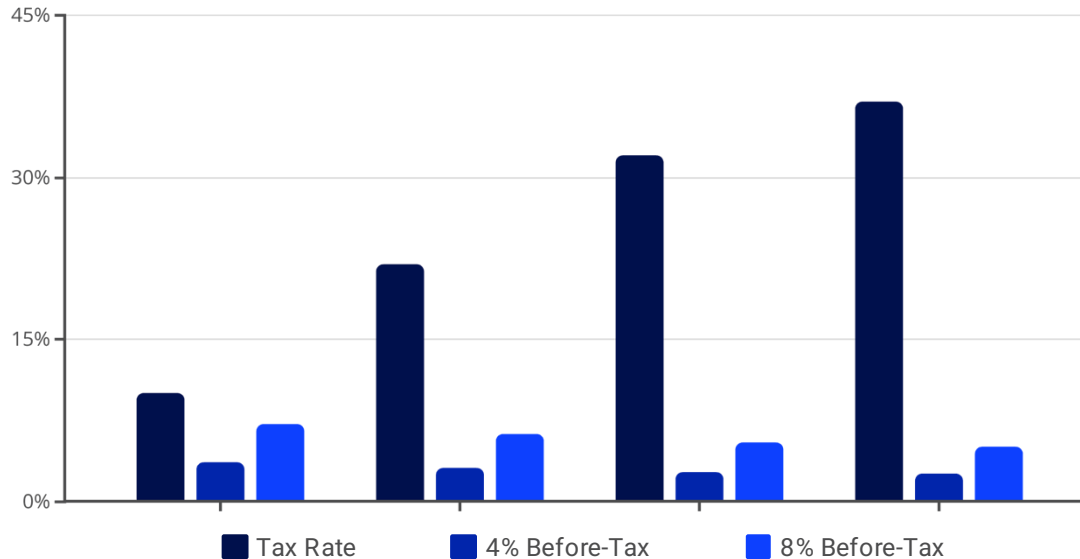


Exhibit 3.2 presents after-tax yields based on a variety of tax rates and before-tax yields. For example, a taxable security with a before-tax yield of 4 percent will generate an after-tax yield of 3.60 percent to an investor in the 10 percent tax bracket, 3.12 percent to an investor in the 22 percent tax bracket, and so on. This exhibit shows why investors in high tax brackets are attracted to tax-exempt securities.

Exhibit 3.2 After-Tax Yields Based on Various Tax Rates and Before-Tax Yields

BEFORE-TAX YIELD				
TAX RATE	2 %	4 %	6 %	8 %
10%	1.80%	3.60%	5.40%	7.20%
22	1.56	3.12	4.68	6.24
32	1.36	2.72	4.08	5.44
37	1.26	2.52	3.78	5.04

After-Tax Yield Comparison



Tax Impact on Returns

Investors in higher tax brackets benefit most from tax-exempt securities. A taxable security with 8% before-tax yield delivers only 6.4% after-tax to someone in the 20% bracket.

To match an 8% tax-exempt yield, a taxable security must offer 10% before-tax yield for the same investor: $Y_{bt} = Y_{at} / (1 - T)$

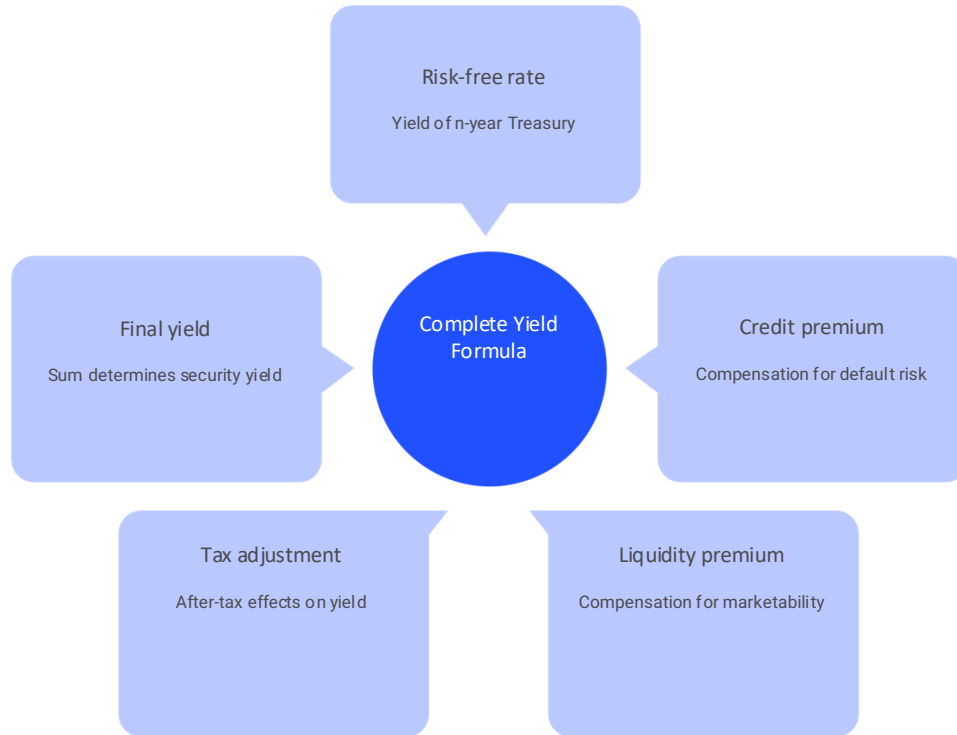
$$Y_{bt} = \frac{Y_{at}}{1 - T}$$

04

Modeling the Yield to be Offered on a Debt Security



The Yield Model Formula



Complete Yield Formula

$$Y_n = R_{f,n} + CP + LP + TA$$

Where:

- Y_n = Yield of n-year debt security
- $R_{f,n}$ = Yield of n-year Treasury security
- CP = Credit risk premium
- LP = Liquidity premium
- TA = Tax status adjustment

The term structure of interest rates defines the relationship between maturity and yield, illustrated by the Treasury yield curve.

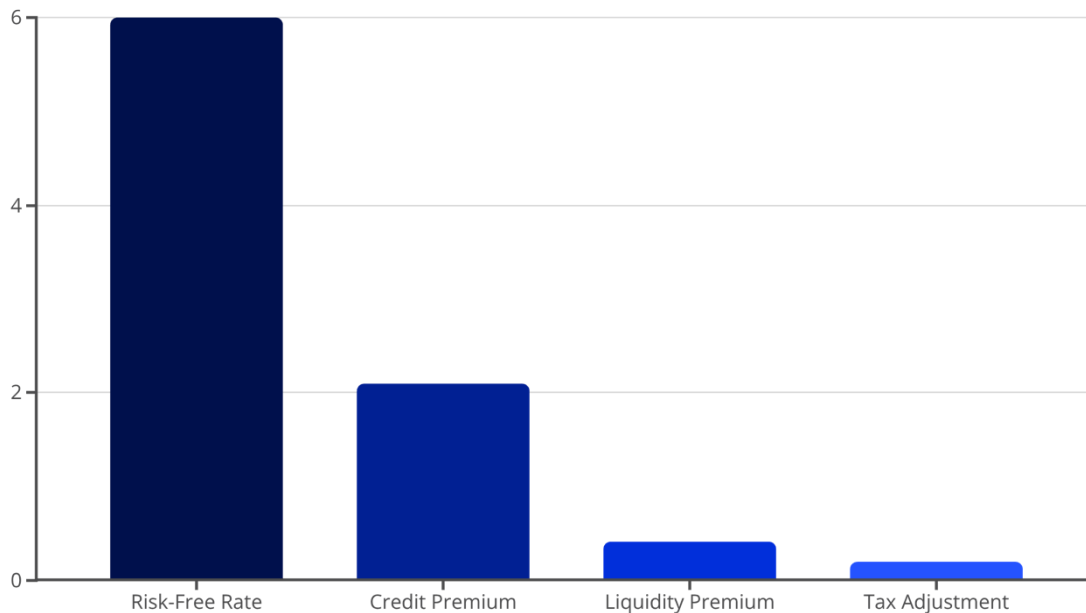
This model helps determine the appropriate yield when issuing debt securities.

CASE STUDY

Elizabeth Co. Bond Pricing Example

Elizabeth Co. plans to issue 10-year bonds and must determine the appropriate yield to attract investors. The company starts with the risk-free Treasury rate and adds premiums based on its specific characteristics.

The calculation demonstrates how corporations systematically build their required yield by accounting for various risk factors that investors demand compensation for.



The Yield Calculation Formula

The appropriate yield for Elizabeth Co.'s bonds combines the risk-free rate with three key premiums that reflect the company's specific risk profile:

$$Y_n = R_{f,n} + DP + LP + TA$$

$$Y_n = 6\% + 2.1\% + 0.4\% + 0.2\% = 8.7\%$$

Risk-Free Rate (6%)

Based on 10-year Treasury securities, representing the baseline return with no default risk.

Credit Risk Premium (2.1%)

Compensation for the possibility that Elizabeth Co. might default on its obligations.

Liquidity Premium (0.4%)

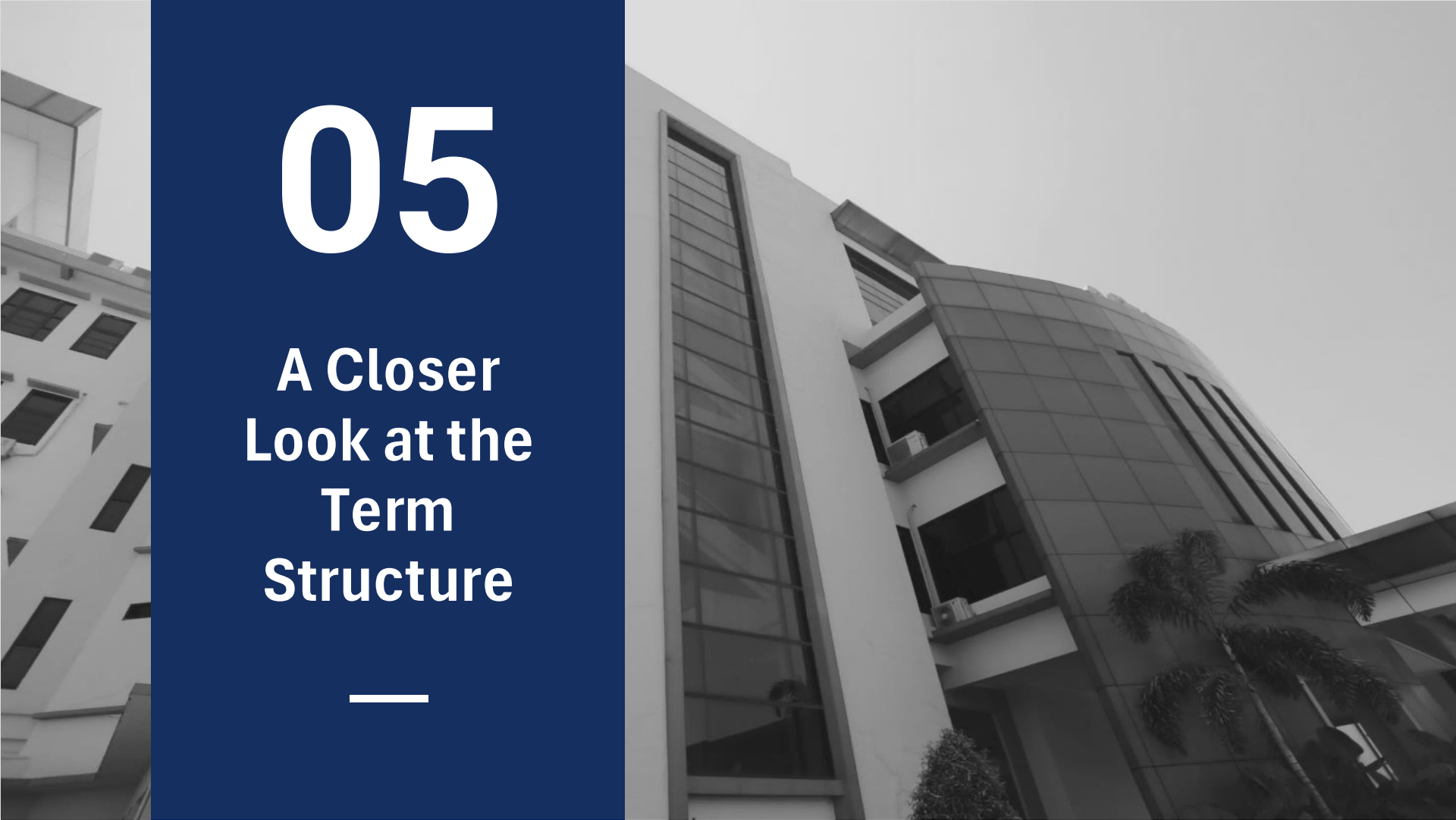
Additional yield to compensate for the difficulty of quickly selling the bonds without loss.

Tax Adjustment (0.2%)

Reflects the tax treatment differences between corporate and government bonds.

05

A Closer Look at the Term Structure





Dynamic Nature of Bond Yields

The appropriate yield for debt securities constantly changes due to fluctuations in multiple factors. The risk-free rate, credit premium, liquidity premium, and tax adjustments all vary with market conditions and economic cycles.

Corporations often consider postponing debt issuance until economic conditions improve and credit risk premiums decrease. However, this strategy carries risk—any savings from a smaller credit premium could be offset by increases in the risk-free rate. Market timing requires careful analysis of all yield components.



Understanding the Term Structure of Interest Rates

Pure Expectations Theory

Term structure determined solely by interest rate expectations

Liquidity Premium Theory

Investors demand premiums for holding less liquid long-term securities

Market Segmentation

Different maturity markets operate independently based on investor preferences

The relationship between term to maturity and annualized yield is complex and requires deeper analysis. Three major theories explain how maturity affects yields, each offering unique insights into market behavior and investor decision-making.

Understanding the Term Structure of Interest Rates



Of all factors affecting debt security yields, term to maturity is the most complex to understand. The relationship between maturity and annualized yield—called the term structure of interest rates—requires deeper examination.

Three major theories explain this relationship, each offering unique insights into how market forces shape the yield curve and influence investment decisions.



Pure Expectations Theory

Pure expectations theory states that the term structure of interest rates is determined solely by market expectations of future interest rates. This theory provides a framework for understanding how investor and borrower behavior shapes the yield curve.

1

Investor Expectations

Investors choose maturities based on their interest rate forecasts, seeking to maximize returns over their investment horizon.

2

Borrower Preferences

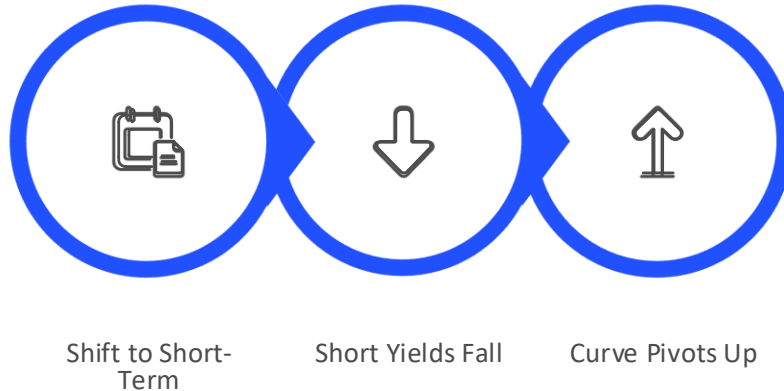
Borrowers select debt maturities to minimize costs based on their rate expectations, influencing demand across the curve.

3

Market Equilibrium

The interaction of these preferences determines yields at each maturity, creating the observable yield curve shape.

Impact of Expected Rising Interest Rates



Market Dynamics

The large supply of funds in the short-term market forces annualized yields down, while reduced supply of long-term funds forces long-term yields up. This creates an upward-sloping yield curve.

Investors accept lower short-term yields because they expect to reinvest at higher rates when securities mature.



Impact of Expected Declining Interest Rates

Investor Behavior: Lock in High Rates

When investors anticipate a decline in interest rates, they shift their preference towards long-term securities. This strategy allows them to secure current higher yields before rates drop, making long-term investments more attractive.

Borrower Strategy: Refinance Later

Conversely, borrowers favor short-term funds. This enables them to access capital now and then refinance their debt at potentially lower interest rates once the expected decline materializes, optimizing their borrowing costs.

Market Result: Yield Curve Shifts

These combined behaviors lead to a reduced supply of funds in the short-term market and an increased supply in the long-term market. This dynamic places upward pressure on short-term yields and downward pressure on long-term yields, causing the yield curve to flatten or invert.



The Forward Rate Concept

Investors can either purchase a security matching their investment horizon or buy shorter-term securities and reinvest at maturity. The market adjusts so expected returns are similar regardless of strategy chosen.

This relationship means a two-year security should offer returns similar to two consecutive one-year investments, and a four-year security should match four consecutive one-year investments.

$$(1 + {}_t i_2)^2 = (1 + {}_t i_1)(1 + {}_{t+1} r_1)$$

where

${}_t i_2$ = known annualized interest rate of a two-year security as of time t

${}_t i_1$ = known annualized interest rate of a one-year security as of time t

${}_{t+1} r_1$ = one-year interest rate that is anticipated as of time $t + 1$ (one year ahead)

CALCULATION EXAMPLE

Calculating the Forward Rate

The forward rate can be isolated by rearranging the pure expectations equation. This rate represents the market's forecast of future interest rates.

$$1 + {}_{t+1}r_1 = \frac{(1 + {}_tj_2)^2}{1 + {}_tj_1}$$
$${}_{t+1}r_1 = \frac{(1 + {}_tj_2)^2}{1 + {}_tj_1} - 1$$

Given a two-year rate of 10% and a one-year rate of 8%, we can calculate the implied one-year forward rate beginning one year from now.

Numerical Solution

$$r_{t+1} = \frac{(1 + 0.10)^2}{1 + 0.08} - 1 = 0.1203704$$

The forward rate of approximately 12.037% represents the one-year rate that must exist one year from now for consecutive one-year investments to match the two-year investment return.



Interpreting Forward Rates

The forward rate serves as an approximation of the market's consensus interest rate forecast. If the market held a different view, supply and demand for securities of different maturities would adjust to capitalize on that information.

1

Forward Rate as Forecast

Market participants use forward rates to gauge expected future conditions and make investment decisions.

2

Market Efficiency

If forward rates are unbiased estimators, the information they contain cannot generate abnormal returns.

3

Self-Correcting Mechanism

Any bias in forward rates would trigger trading that eliminates the bias through price adjustments.

Yield Curve Shapes and Expectations

If the term structure is solely influenced by interest rate expectations, the relationship between forward rates and current rates determines the yield curve shape:



Upward Sloping Curve

Forward rate exceeds today's rate ($r_{t+1} > i_t$), indicating expectations of rising interest rates.



Flat Curve

Forward rate equals today's rate ($r_{t+1} = i_t$), suggesting stable interest rate expectations.



Downward Sloping Curve

Forward rate is below today's rate ($r_{t+1} < i_t$), reflecting expectations of declining rates.

SCENARIO	STRUCTURE OF YIELD CURVE	EXPECTATIONS ABOUT THE FUTURE INTEREST RATE
1. ${}_{t+1}r_1 > {}_t i_1$	Upward slope	Higher than today's rate
2. ${}_{t+1}r_1 = {}_t i_1$	Flat	Same as today's rate
3. ${}_{t+1}r_1 < {}_t i_1$	Downward slope	Lower than today's rate



Extended Forward Rate Calculations

Forward rates can be determined for any maturity. The pure expectations theory equality for a three-year horizon extends the two-year relationship:

$$(1 + {}_t i_3)^3 = (1 + {}_t i_1)(1 + {}_{t+1} r_1)(1 + {}_{t+2} r_1)$$

where

${}_t i_3$ = annualized interest rate of a three-year security as of time t

${}_{t+2} r_1$ = one-year interest rate that is anticipated as of time $t + 2$
(two years ahead)

Rearranging to isolate the forward rate:

$$1 + {}_{t+2} r_1 = \frac{(1 + {}_t i_3)^3}{(1 + {}_t i_1)(1 + {}_{t+1} r_1)}$$
$${}_{t+2} r_1 = \frac{(1 + {}_t i_3)^3}{(1 + {}_t i_1)(1 + {}_{t+1} r_1)} - 1$$

This allows estimation of interest rates further into the future by combining known current rates with previously calculated forward rates.

ADVANCED EXAMPLE

Multi-Period Forward Rate Example

Given Information

- One-year rate: 8%
- Three-year rate: 11%
- Previously calculated one-year forward rate (one year ahead): 12.037%

Calculation

$$\begin{aligned} {}_{t+2}r_1 &= \frac{(1 + {}_t i_3)^3}{(1 + {}_t i_1)(1 + {}_{t+1}r_1)} - 1 \\ &= \frac{(1 + 0.11)^3}{(1 + 0.08)(1 + 0.12037)} - 1 \\ &= \frac{1.367631}{1.21} - 1 \\ &= 13.02736\% \end{aligned}$$

Interpretation

The market anticipates that two years from now, the one-year interest rate will be approximately 13.03%.

This demonstrates how the yield curve embeds expectations about future rate movements across multiple time periods.



Forecasting Multi-Year Forward Rates

The yield curve can forecast annualized rates for periods longer than one year. According to pure expectations theory, a one-year investment followed by a two-year investment should offer the same annualized yield as a three-year security purchased today:

$$(1 + {}_t i_3)^3 = (1 + {}_t i_1)(1 + {}_{t+1} r_2)^2$$

Where ${}_{t+1} r_2$ is the two-year rate anticipated one year ahead. Rearranging:

$$({}_{t+1} r_2)^2 = \frac{(1 + {}_t i_3)^3}{1 + {}_t i_1}$$

Using the previous example with one-year rate of 8% and three-year rate of 11%, the anticipated two-year rate beginning one year from now is approximately 12.53%.

ADVANCED EXAMPLE

Recall that today's annualized yields for one-year and three-year securities are 8 percent and 11 percent, respectively. With this information, ${}_{t+1}r_2$ is estimated as follows:

$$\begin{aligned}(1 + {}_{t+1}r_2)^2 &= \frac{(1 + {}_t i_3)^3}{1 + {}_t i_1} \\ &= \frac{(1 + 0.11)^3}{1 + 0.08} \\ &= 1.266325 \\ 1 + {}_{t+1}r_2 &= \sqrt{1.266325} \\ &= 1.1253 \\ {}_{t+1}r_2 &= 0.1253\end{aligned}$$

Thus, the market anticipates an annualized interest rate of about 12.53 percent for two-year securities beginning one year from now.



Market Efficiency and Forward Rates

Pure expectations theory assumes forward rates are unbiased estimators of future interest rates. If this holds true, financial market efficiency is supported, and information implied by market rates cannot generate abnormal returns.

When new information arrives, investor preferences change, yields adjust, and implied forward rates adjust accordingly. Any attempt to capitalize on perceived forward rate bias would trigger market movements that eliminate the bias through price and yield adjustments.



Stability of Long-Term Rates

The Averaging Effect

As expectations about consecutive short-term rates change over time, the average of these rates exhibits less volatility than individual short-term rates.

This mathematical relationship explains why long-term rates tend to be much more stable than short-term rates, even when short-term rate expectations fluctuate significantly.

Short-term Rates

Fluctuate significantly over brief periods

Long-term Rates

Geometric average yields greater stability



ALTERNATIVE THEORY

Liquidity Premium Theory

Liquidity premium theory offers an alternative explanation for the term structure of interest rates. Some investors prefer short-term over long-term securities because shorter maturity represents greater liquidity.

01

Liquidity Preference

Investors may only hold long-term securities if compensated with a premium for reduced liquidity.

02

Price Sensitivity

Long-term securities can be liquidated before maturity, but their prices are more sensitive to interest rate movements.

03

Liquidity Advantage

Short-term securities are more liquid because they can be converted to cash without significant loss in value.



Calculating Forward Rates with Liquidity Premium

When expectations theory combines with liquidity theory, the yield on a security differs from consecutive shorter-term investments. The two-year security yield is determined as:

$$(1 + {}_t i_2)^2 = (1 + {}_t i_1)(1 + {}_{t+1} r_1) + LP_2$$

The forward rate formula accounting for liquidity premium becomes:

$${}_{t+1} r_1 = \frac{(1 + {}_t i_2)^2}{1 + {}_t i_1} - 1 - \frac{LP_2}{1 + {}_t i_1}$$

Example Calculation

Reconsider the example where $i_1 = 8$ percent, and $i_2 = 10$ percent, and assume that the liquidity premium on a two-year security is 0.5 percent. The one-year forward rate can then be derived from this information as follows:

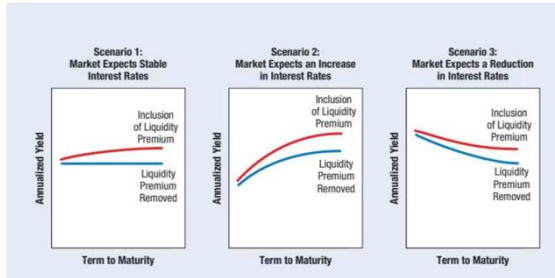
Key Insight

Forecasts of future interest rates implied by yield curves are reduced slightly when accounting for the liquidity premium, providing more accurate expectations.

This estimate of the one-year forward rate is lower than the estimate derived in the previous related example in which the liquidity premium was not considered. The previous estimate (12.037 percent) of the forward rate probably overstates the market's expected interest rate because it did not account for a liquidity premium.

Impact of Liquidity Premium Under Different Scenarios

The exhibit shows how liquidity premium affects yield curves under three market expectations:



01

Stable Rates Expected

Blue line is flat; red line curves upward due to liquidity premium

02

Rising Rates Expected

Both lines slope upward; red line steeper with premium included

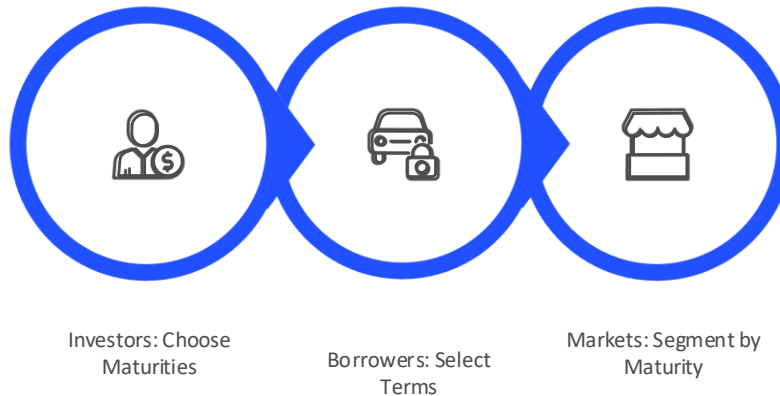
03

Falling Rates Expected

Both lines slope downward; red line above blue throughout

THEORY #2

Segmented Markets Theory



Market Participants

- Pension funds and life insurance companies prefer long-term investments matching long-term liabilities
- Commercial banks prefer short-term investments coinciding with short-term liabilities
- Corporations choose maturities based on funding needs rather than rate expectations



How Market Imbalances Shape the Yield Curve

1

Scenario 1: Upward Slope

Most investors have short-term funds; most borrowers need long-term funds. Result: downward pressure on short-term yields, upward pressure on long-term yields.

2

Scenario 2: Downward Slope

Most investors seek long-term investments; most borrowers need short-term funds. Result: upward pressure on short-term yields, downward pressure on long-term yields.

3

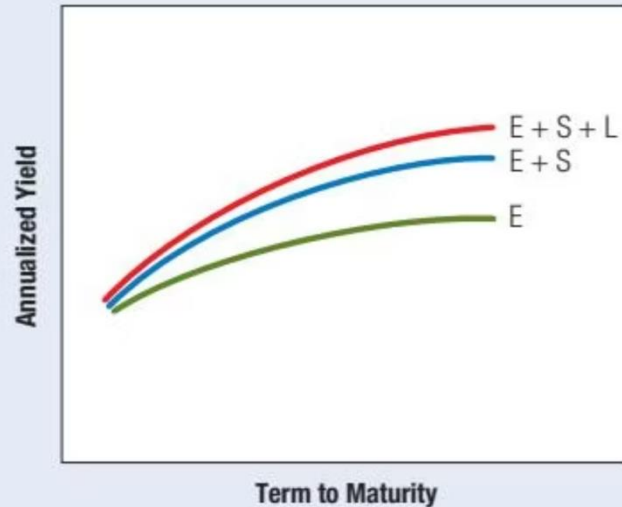
Balanced Markets

When supply and demand are balanced between short-term and long-term markets, yields become more similar across maturities.

❏ **Preferred Habitat Theory** offers a compromise: investors and borrowers normally concentrate on particular maturity markets, but certain events may cause them to venture from their natural markets. This acknowledges that both maturity preferences and interest rate expectations influence the yield curve.

Integrating All Three Theories

Research suggests that expectations theory, liquidity premium theory, and segmented markets theory all have validity. Understanding how they work together provides the most complete picture of yield curve dynamics.



Combined Effects Example

Assume three conditions exist simultaneously:

1. Investors expect interest rates to rise (expectations theory)
2. Most borrowers need long-term funds while investors have short-term funds (segmented markets)
3. Investors prefer more liquidity (liquidity premium)

Curve E shows expectations effect alone. Curve E+S adds segmented markets impact. Curve E+S+L includes all three effects, creating the steepest upward slope.

Practical Applications of the Term Structure



Forecasting Interest Rates

The yield curve's shape assesses general expectations about future rates. Upward-sloping curves suggest higher rates ahead; downward-sloping curves indicate lower rates expected.



Forecasting Recessions

Flat or inverted yield curves often indicate recession ahead. In March 2007, a slight negative slope preceded the December 2007 recession and 2008-2009 credit crisis.



Investment Decisions

Investors may "ride the yield curve" by purchasing longer-term securities for higher yields even with short-term horizons, selling in the secondary market when needed.



Financing Decisions

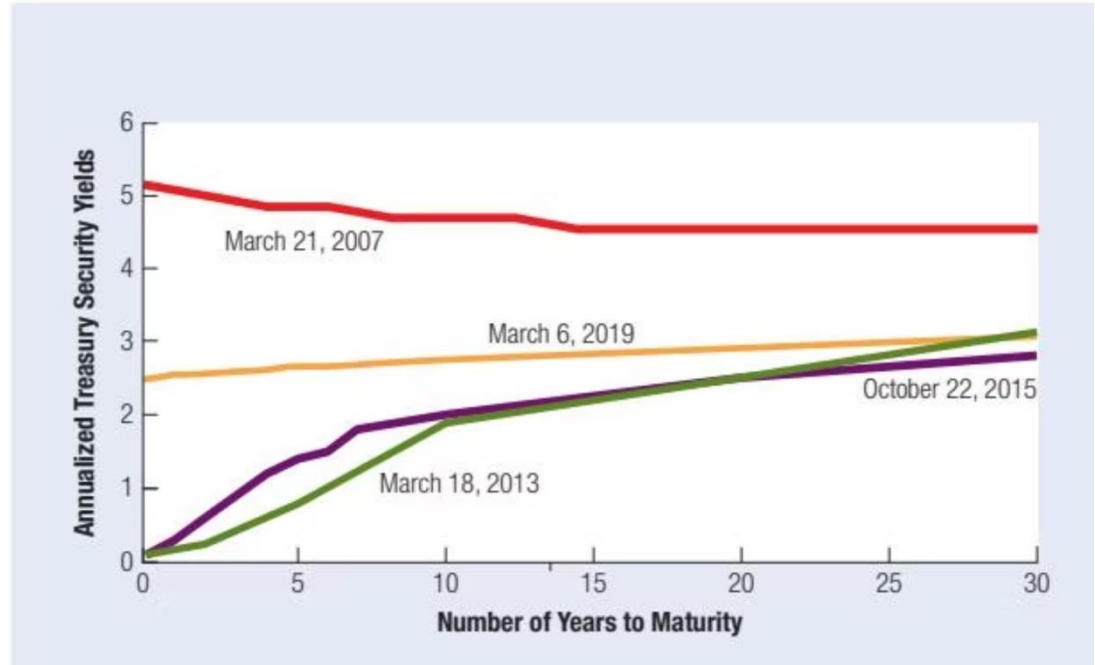
Firms use yield curves to time borrowing. An upward-sloping curve may encourage short-term borrowing with plans to refinance later at potentially lower rates.

Historical Yield Curve Evolution

Key Observations

Yield curves are usually upward sloping, but occasional downward slopes occur. Recent years have shown very low yield curves, reflecting low annualized interest rates across all maturities.

The March 21, 2007 curve exhibited a slight downward slope—a warning sign that preceded the financial crisis. By contrast, the October 22, 2015 and March 6, 2019 curves showed more typical upward slopes, though at historically low levels.



Key Takeaways



International Variation

Each country has different yield curve shapes based on its own supply and demand for loanable funds, though movements across countries are positively correlated due to integrated financial markets.



Multiple Factors

The term structure depends on interest rate expectations, investor preferences for liquidity, and unique needs of investors and borrowers in each maturity market.



Practical Tool

Understanding yield curve dynamics helps forecast rates and recessions, and guides investment and financing decisions for both institutions and corporations.

Should a yield curve influence a borrower's preferred loan maturity? An upward-sloping curve suggests lower short-term rates, but also signals potential rate increases ahead. Borrowers must weigh immediate savings against future refinancing uncertainty.



Thank you